Banksian Fibron Space

Reciprocal Winding and Hierarchical Geometric Construction

Working edition · November 9, 2026

Abstract

Banksian Fibron Space constructs spatial trajectories by attaching a local waveform to an anchor with ordered rotations. A 120° circular arc and its reflection form a lens whose perimeter is two-thirds of the reference circumference. Its normalized traversal yields reciprocal phase factors 3/2 and 2/3. We assign these factors to rotation axes to define the Wave-Fold and Shear-Fold. We derive their length bounds, the symmetry of a paired shear about its moving midpoint, and the conditions for recurrence through a finite hierarchy. For the canonical rates, the minimum simultaneous phase-return interval is 12π. Translating the anchor during that interval produces a spatial thread whose internal configuration repeats along its path.

Introduction

Consider a point tracing an ellipse on a sheet. If the sheet turns about two axes while the point moves, an observer attached to the sheet still sees the ellipse. An observer in the surrounding space sees a curve with three changing coordinates. Both observers describe the same moving point. Banksian Fibron Space specifies the winding rates, the order of those turns, and how to attach another frame to any point already constructed.

We use *anchor* for the origin and oriented axes that carry a local construction. A *Fibron* is the geometric traversal component of a folded waveform over its cycle. Throughout, coordinates are dimensionless and t denotes progression. Three questions guide the construction. Where do the winding factors come from? What does a fold preserve? Under what conditions does a point return when the frames carrying it also move?

1. Construct the radial lens · BFS-01–09

Consider two circles of radius $R > 0$ with centers $O = (0, 0)$ and $L = (-R, 0)$. Subtraction of the two circle equations gives $x=-R/2$ at their intersections. Substitution gives U and D at $y=\pm\sqrt{3}R/2$. The chord UD therefore has length $\sqrt{3}R$. Each endpoint forms an equilateral triangle with the two centers.

Reflect the left 120° arc of the circle centered at O across UD by sending (x,y) to $(-R-x,y)$. The two boundary arcs each have length $2\pi R/3$. Their combined length is $4\pi R/3$, or two-thirds of a full circumference. If s measures distance along the boundary, the normalized boundary phase $\tau=2\pi s/(4\pi R/3)$ equals $(3/2)\alpha$, where $\alpha=s/R$. The inverse conversion is $\alpha=(2/3)\tau$. At one complete traversal, τ has advanced by 2π while α has advanced by $4\pi/3$. We use their conversion factors as the two angular rates in the spatial construction. At each endpoint the boundary changes arcs, so this distance correspondence alone does not specify the coordinates of a folded curve.

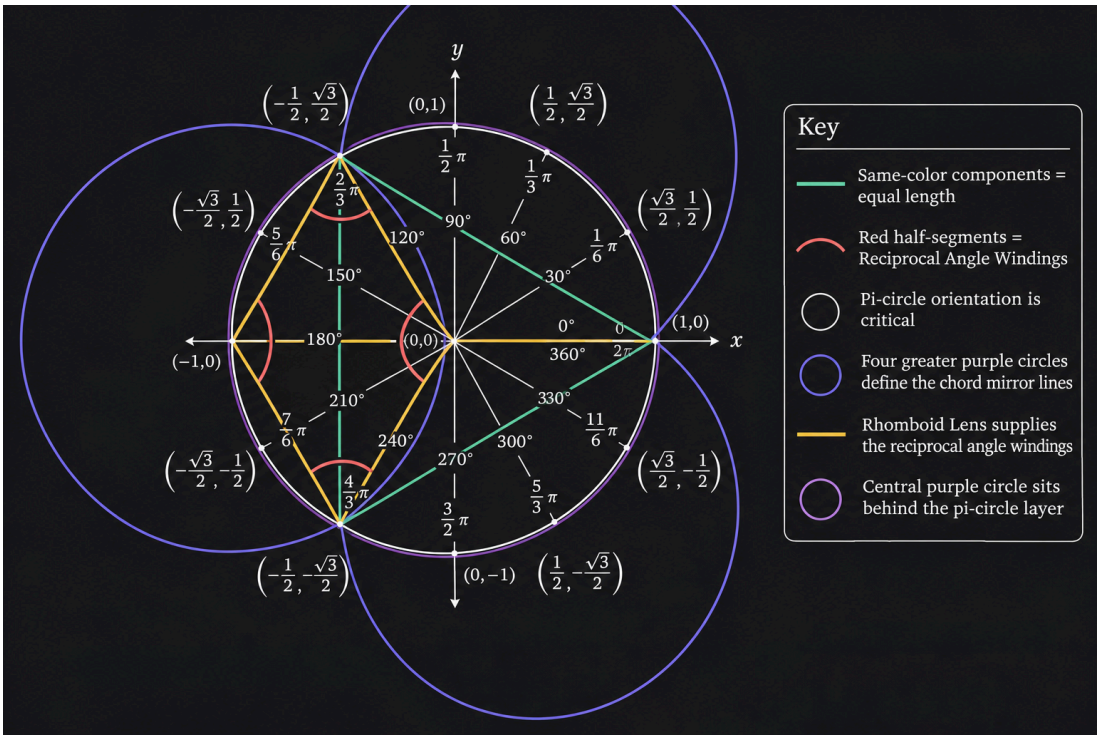


Figure 1 · Trigonometric chord mirror construction. The yellow rhomboid joins the circle centers and the two endpoints of their common chord. The red angle marks at its vertices identify the reciprocal winding angles. The arc-length calculation above gives the numerical phase conversion used by the folding operators.

2. Define the Wave-Fold · BFS-10–14

The construction designates three rhomboid lenses per reference circle. A Wave-Fold can use one, two, or all three. The number used describes how the same wave is expressed across the construction. This is a convention of the framework. The two-axis operator below is the instance worked out in the proof register.

Take R>0, S≠0 and a real progression parameter t. Angles are in radians. The winding rates S/2 and 2/S have product one. For the canonical choice S=3, write a=3t/2 and b=2t/3. With A=R√3/2 and B=2/(R√3), we also have AB=1. The local waveform w=(A cos t, −B sin t, 0) traces an ellipse of area π.

With column vectors, define QW=Rz(b)Ry(a). Applying the rightmost rotation first gives

$$W(t) = \begin{pmatrix} A\cos t\cos a\cos b + B\sin t\sin b \\ A\cos t\cos a\sin b - B\sin t\cos b \\ -A\cos t\sin a \end{pmatrix}.$$

Both factors are proper orthogonal matrices, so $|W|^2=A^2\cos^2t+B^2\sin^2t$. The point stays between distances min(A,B) and max(A,B) from its anchor. Rotation preserves the instantaneous distance, even as the ellipse makes that distance vary with t.

To compare the two supplied Wave-Fold bases, let $C = 2\sqrt{3}/R$. In the two-lens base the local point is (C cos t, −A sin t, 0) and its frame applies Rz(b)Ry(a). In the three-lens base the same point gains a local Z component C cos t, and the frame applies Rz(b)Ry(a)Rx(b). Both express a single wave. The Shear-Fold developed below introduces a pair of poles and derives their nucleus. At the saved settings R=S=3, both Wave-Fold bases share the canonical 12π phase return. The ellipse-area identity above uses the reciprocal amplitudes A and B defined in BFS-11.

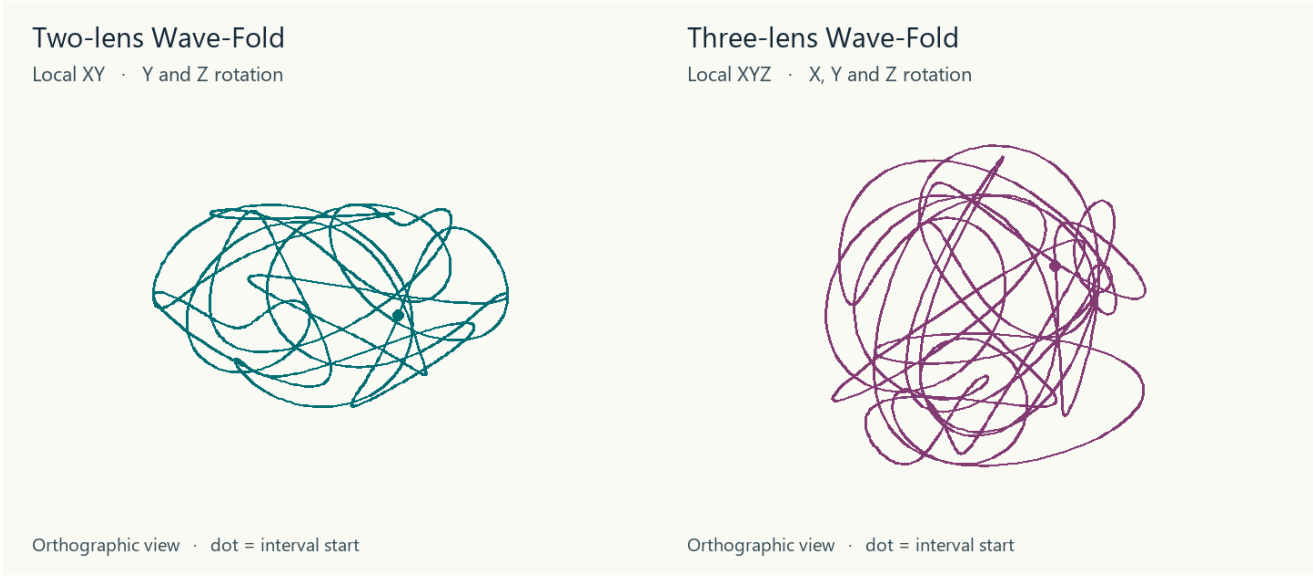


Figure 2 · Two expressions of a Wave-Fold. The left and right panels use the formulas in Base2Lens Wave-Fold.json and 3Lens Wave-Fold.json, respectively. Both have R=S=3, a stationary anchor, and 1,441 sample points from 0 through 12π, including both endpoints. The panels share a projection and scale. Two traces can cross in this projection while lying at different depths.

The distinction between a family and a particular operator matters even in the local plane. The registered ellipse has semiaxes A and B and area π. The supplied two-lens ellipse has semiaxes C and A. Since CA=3, its area is 3π. Both points follow the same pair of winding rates. A measurement of local area distinguishes the two choices before any rotation is applied.

3. Construct the poles and nucleus · BFS-13C–E, BFS-17–21A

For the Shear-Fold set QS=Rz(b)Ry(a)Rx(b). Define M=(−B cos t,0,0) and D=(0,A sin t,−B cos t). The local poles p_+=M+D and p_-=M−D are related by a half-turn about local X. Their midpoint is M. For an anchor o, the world midpoint is N=o+QS M, and the world poles are N±QS D. Their distances from N equal $\sqrt{(A^2\sin^2 t+B^2\cos^2 t)}$, which is strictly positive. This gives a measurable separation at every phase.

At t=0 the local poles are (−B,0,−B) and (−B,0,B). Their nucleus lies between them at (−B,0,0). A quarter-cycle of the local waveform later, at t=π/2, the nucleus passes through the anchor while the poles occupy (0,A,0) and (0,−A,0) in the local frame. Thus even when the nucleus reaches the anchor, the two poles remain separated by 2A. This example also specifies how to construct a midpoint in software. Average the two local poles and transform the result once, or average the two world positions directly. Both procedures give N.

4. Recurrence, progression and threads · BFS-15–16, BFS-22, BFS-27

A common return of the three phases at rates 1, 3/2 and 2/3 requires T=2πn. The second rate requires even n and the third requires n divisible by three. Their first common return is n=6. Thus T=12π restores both canonical matrices and every local waveform sharing that period. A stationary anchor gives world-space recurrence. A projection can have a shorter period.

For the Fold Progression $x_a(t)$=2t/R, the anchor advances by 24π/R each canonical interval. Over consecutive intervals the same internal configuration appears at translated positions. The trajectory obtained by advancing the anchor is the thread. For general S, simultaneous phase recurrence requires rational commensurability with the base rate.

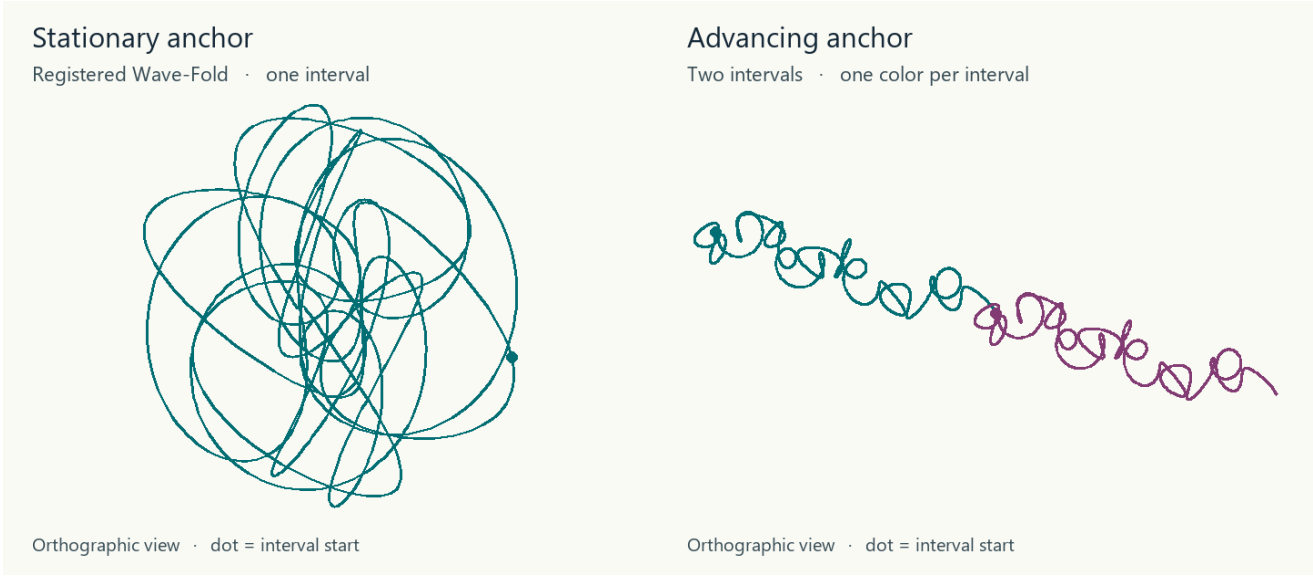


Figure 3 · Fold progression. Both panels use the registered Wave-Fold with R=S=3 and 1,440 equal sampling intervals per 12π. The stationary fold returns to its starting point. In the right panel, the anchor advances 8π along X per interval. The second color begins where the first ends. Both panels use the same viewing direction, with separate scales to fit the translating trajectory.

5. Compose a hierarchy · BFS-23–26

Any point with an assigned frame can serve as an anchor for a descendant, including a pole or a derived midpoint. Let P_p and H_p be the parent's world position and orientation. A child with local displacement u and local rotation Q has world position $P_p + H_p u$ and world orientation $H_p Q$. Repeated application gives an ordered product of orthogonal matrices. If the root and every local displacement and rotation share period T, recurrence passes from each parent to its children by induction. Swapping the two canonical rates at a nested level preserves their common return.

For example, the supplied Hierarchical Complex places HFCa in the frame of HFAnchor, attaches Shear Frame to HFCa, and places A and Am in Shear Frame. A change to HFAnchor therefore carries both poles. A change to the coordinates of A moves that pole and changes the derived midpoint. The order matters at each attachment. If a parent turns before its child translates, the translation follows the parent's turned axes.

6. Experimental Shear Division and scope · BFS-26, BFS-28–31

A shared anchor can carry several waveform points. Their centroid is their arithmetic mean and transforms with their common frame. The supplied Shear-Division experiment averages three world positions. Its coefficients and winding assignments specify one experiment. A general rule for allocating amplitudes and orientations among N participants would extend that example.

7. Working with the scope

On first use, Fibron Scope opens with the registered Wave-Fold. To reproduce the phase return, select its wave point, open Matrices, and compare one 12π interval. The reported displacement and orientation differences measure agreement at the two evaluated endpoints. The period argument in Section 4 establishes why that agreement extends to every starting phase for the canonical construction.

To reproduce the translating example, select the wave's anchor under Objects and enable its X progression. Compare the wave again. Across the interval, the change in the wave's anchor-relative position is zero within numerical precision. Its world displacement has magnitude $24\pi/R$. The Next 12π control carries the same progression into the next viewing window. Sample density and connected versus discrete rendering can be changed under Parameters.

The five archived scenes are available under Presets. The two Wave-Fold scenes supply the constructions in Figure 2. Shear Fold and Hierarchical Complex preserve the recorded pole formulas and frame attachments. The registered Shear-Fold under Objects directly implements the paired-pole equations of Section 3. The ShearDivision.json preset preserves the three-contributor experiment with its specified coefficients. The scope also offers an exploratory generator with evenly spaced phases on a common anchor. The generator's formulas allow the phase spacing to be changed. Centroid covariance holds for any chosen set of contributors in a common frame.

8. Limits and open questions

The lens perimeter fixes a conversion between two measures of traversal. Assigning those factors to particular axes introduces a spatial definition. An exact identification of a projected folded curve with the circular lens would require a coordinate correspondence. Likewise, a common phase return establishes a repeat interval while leaving the shortest period of a particular spatial projection to be determined.

The construction supports prescribed anchor motion and finite frame hierarchies. Within the geometric setting, the formulas already give three direct checks on a rendering: the registered Wave-Fold's distance bound, the Shear-Fold's pole separation, and the return of a periodic hierarchy. Each can be evaluated independently of camera angle and sampling density.

Proof index

The complete mathematical proof register follows in the publication appendix. Entries marked [P] give established properties, [C] conditional theorems, [D] definitions, and [O] open correspondences.

Appendix · Mathematical proof register

Banksian Fibron Space

Mathematical Proof Register

Revised November 9, 2026

Entries are cited by their BFS identifiers. Each entry records its mathematical status.

Status:
[P] Exact identity or proved property
[C] Conditional theorem
[D] Definition
[O] Open correspondence

Conventions: R > 0, S ≠ 0, t ∈ ℝ. Angular arguments use radians. All geometry is dimensionless until physical units are assigned. Rotation matrices act on column vectors, with the rightmost transformation applied first.

The rotation multiplication order below is the **declared mathematical convention**. FieldLab uses the same column-vector convention, with local rotation order $R_z R_y R_x$.

A. Circle, chord, and lens

BFS-01 [P] · Circle intersections

For equal-radius circles with centers O=(0,0) and L=(−R,0):

$$U = (-R/2, \sqrt{3}\,R/2), \qquad D = (-R/2, -\sqrt{3}\,R/2).$$

Both points satisfy both circle equations.

BFS-02 [P] · Equilateral triangle complex

Triangles LUO and LDO are equilateral, with every side equal to R. Their shared construction provides the 60°, 120°, and bisected 30°–60°–90° relationships.

BFS-03 [P] · 120° chord

$$c = 2R\sin(\pi/3) = R\sqrt{3}.$$

Thus UD=R√3.

BFS-04 [P] · Component right triangle

$$(R/2)^2 + (\sqrt{3}\,R/2)^2 = R^2.$$

The component hypotenuse is R. The full chord is R√3.

BFS-05 [P] · Circular-lens reflection

Reflect the left 120° circle arc across the chord line x=−R/2 using:

$$F(x, y) = (-R - x, y).$$

The original and reflected arcs form an exact circular lens. Both have radius R and meet at U and D.

BFS-06 [P] · Lens perimeter ratio

$$L_{\text{lens}} = 2(2\pi R/3) = 4\pi R/3.$$

With reference circumference C=2πR:

$$L_{\text{lens}}/C = 2/3, \qquad C/L_{\text{lens}} = 3/2.$$

BFS-07 [P] · Normalized lens phase

Let s be cumulative distance traveled along the lens boundary, τ=2πs/L_lens, and α=s/R.

Then:

$$\tau = (3/2)\alpha, \qquad \alpha = (2/3)\tau.$$

This is a continuous arc-length correspondence, with piecewise orientation at the two lens endpoints.

BFS-08 [P] · Lens area

$$A_{\text{lens}} = R^2(2\pi/3 - \sqrt{3}/2).$$

Each boundary arc encloses a segment of area $\pi R^2/3 - \sqrt{3}R^2/4$ between the arc and the chord. Adding the two segment areas gives the stated lens area.

BFS-09 [P] · Distinction from the 90° construction

For the same circle radius:

$$c_{90} = R\sqrt{2}, \qquad c_{120} = R\sqrt{3}.$$

The respective arc-to-chord ratios are $\pi/(2\sqrt{2})$ and $2\pi/(3\sqrt{3})$, obtained by dividing each arc length by its corresponding chord.

B. Reciprocal basis and Wave-Fold

BFS-10 [D/P] · Reciprocal winding basis

Define:

$$g_+(S) = S/2, \qquad g_-(S) = 2/S.$$

Their product is identically 1.

At S=3:

$$g_+ = 3/2, \qquad g_- = 2/3.$$

BFS-11 [P] · Reciprocal geometric amplitudes

Define:

$$A_R = R\sqrt{3}/2, \qquad B_R = 2/(R\sqrt{3}).$$

Then:

$$A_R B_R = 1.$$

BFS-12 [D/P] · Canonical local Wave-Fold example

Define the local XY waveform:

$$w(t) = \begin{pmatrix} A_R \cos t \\ -B_R \sin t \\ 0 \end{pmatrix}.$$

Its local coordinate identity is:

$$(x/A_R)^2 + (y/B_R)^2 = 1.$$

The unrotated waveform is an ellipse of area πA_RB_R=π. It is not itself the reflected circular lens.

BFS-13 [P] · Fundamental rotation matrices

The three elementary rotations are:

$$R_x(c) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos c & -\sin c \\ 0 & \sin c & \cos c \end{pmatrix}$$

$$R_y(a) = \begin{pmatrix} \cos a & 0 & \sin a \\ 0 & 1 & 0 \\ -\sin a & 0 & \cos a \end{pmatrix}$$

$$R_z(b) = \begin{pmatrix} \cos b & -\sin b & 0 \\ \sin b & \cos b & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives:

$$R_j^{\mathsf{T}} R_j = I, \qquad \det(R_j) = 1$$

for every axis j.

Thus every elementary rotation is orthogonal and preserves vector length.

BFS-13A [P] · Explicit Wave-Fold matrix composition

Set:

$$a = 3t/2, \qquad b = 2t/3.$$

Define:

$$Q_W(t) = R_z(b)R_y(a).$$

Multiplying the matrices:

$$Q_W = \begin{pmatrix} \cos b\cos a & -\sin b & \cos b\sin a \\ \sin b\cos a & \cos b & \sin b\sin a \\ -\sin a & 0 & \cos a \end{pmatrix}$$

This is the complete two-axis Wave-Fold orientation operator under the declared convention.

BFS-13B [P] · Explicit Wave-Fold coordinates

Apply BFS-13A to BFS-12:

$$W(t) = Q_W(t)w(t).$$

The three spatial components are:

$$W_x = A_R \cos t \cos a \cos b + B_R \sin t \sin b$$
$$W_y = A_R \cos t \cos a \sin b - B_R \sin t \cos b$$
$$W_z = -A_R \cos t \sin a$$

Therefore, a local XY waveform produces a three-dimensional trajectory through the ordered Y/Z rotation composition.

No additional waveform dimension needs to be introduced.

BFS-13C [P] · Explicit three-axis Shear-Fold matrix

Use the same reciprocal rates, with X and Z sharing b:

$$a = 3t/2, \qquad b = 2t/3.$$

Define:

$$Q_S(t) = R_z(b)R_y(a)R_x(b)$$

For compactness, let:

$$c_a = \cos a, \quad s_a = \sin a, \quad c_b = \cos b, \quad s_b = \sin b.$$

Multiplying all three matrices gives:

$$Q_S = \begin{pmatrix} c_b c_a & c_b s_b(s_a - 1) & s_b^2 + c_b^2 s_a \\ s_b c_a & c_b^2 + s_b^2 s_a & c_b s_b(s_a - 1) \\ -s_a & c_a s_b & c_a c_b \end{pmatrix}$$

This orientation operator carries both the three-axis Wave-Fold and the canonical Shear-Fold. BFS-17 specifies the paired local poles of the Shear-Fold.

The shared X/Z rate does not make those rotations interchangeable. The order of matrix multiplication remains significant.

BFS-13D [P] · Orthogonality of the complete folds

For Wave-Fold:

$$Q_W^{\mathsf{T}} Q_W = R_y^{\mathsf{T}} R_z^{\mathsf{T}} R_z R_y = I.$$

For Shear-Fold:

$$Q_S^{\mathsf{T}} Q_S = R_x^{\mathsf{T}} R_y^{\mathsf{T}} R_z^{\mathsf{T}} R_z R_y R_x = I.$$

Both have determinant 1.

Consequently:

$$|Q_W v| = |Q_S v| = |v|$$

for any local vector v.

This proves preservation of each instantaneous local vector length under either folding operator.

BFS-13E [P] · Inverse and ordered composition

For either folding matrix Q:

$$Q^{-1} = Q^{\mathsf{T}}$$

Reversing a fold's orientation is therefore achieved by transposing its matrix.

In general:

$$R_z(b)R_y(a) \neq R_y(a)R_z(b).$$

Ordered rotation composition is not commutative. This allows different hierarchical arrangements of identical rotation factors to produce distinct spatial trajectories.

BFS-14 [P] · Waveform boundedness

For BFS-12:

$$|w(t)|^2 = A_R^2 \cos^2 t + B_R^2 \sin^2 t.$$

Thus:

$$\min(A_R, B_R) \leq |w(t)| \leq \max(A_R, B_R).$$

BFS-13D preserves this norm through rotation. Reciprocal amplitudes do not imply a constant radial distance R.

BFS-15 [C] · Canonical 12π phase recurrence

The relevant angular rates are:

$$1, \qquad 3/2, \qquad 2/3.$$

At T=12π, the accumulated angles are:

$$12\pi = 6(2\pi)$$
$$(3/2)(12\pi) = 9(2\pi)$$
$$(2/3)(12\pi) = 4(2\pi).$$

Each phase returns to its initial value modulo 2π.

To prove the minimum common phase return, require the base phase to make n complete revolutions, so T=2πn.

The 3/2 rate requires n to be even. The 2/3 rate requires n to be divisible by 3.

The smallest positive such integer is n=6.

Therefore:

$$T_{\text{common}} = 12\pi$$

This is the minimum simultaneous **phase-return interval**, not necessarily the minimum period of every resulting spatial projection.

BFS-15A [P/C] · Matrix recurrence at 12π

Because the arguments of every constituent rotation advance by integer multiples of 2π:

$$R_x(b(t + 12\pi)) = R_x(b(t))$$
$$R_y(a(t + 12\pi)) = R_y(a(t))$$
$$R_z(b(t + 12\pi)) = R_z(b(t)).$$

Therefore:

$$Q_W(t + 12\pi) = Q_W(t)$$
$$Q_S(t + 12\pi) = Q_S(t)$$

Combining this with the 2π-periodicity of the local waveform establishes:

$$Q(t + 12\pi)w(t + 12\pi) = Q(t)w(t)$$

for either canonical folding operator and any local waveform that repeats after 12π.

BFS-16 [C] · Positional closure

For a world-space point:

$$P(t) = o(t) + Q(t)w(t),$$

if the anchor o, matrix Q, and local waveform w share period T, then:

$$P(t + T) = P(t).$$

For a stationary anchor and either canonical operator, positional recurrence follows at 12π whenever the local waveform repeats after 12π.

C. Shear-Fold, poles, and relational midpoint

BFS-17 [D] · Canonical Shear-Fold pair

Using BFS-11, define:

$$p_+(t) = \begin{pmatrix} -B_R \cos t \\ A_R \sin t \\ -B_R \cos t \end{pmatrix}$$

$$p_-(t) = \begin{pmatrix} -B_R \cos t \\ -A_R \sin t \\ B_R \cos t \end{pmatrix}.$$

These are local geometric projections to which the three-axis matrix of BFS-13C may be applied.

BFS-18 [P] · Axial mirror pairing

$$p_- = \mathrm{diag}(1, -1, -1) p_+.$$

The pair is related by a 180° rotation about its shared local X axis.

BFS-19 [P] · Exact nucleus decomposition

Let:

$$M = (p_+ + p_-)/2, \qquad D = (p_+ - p_-)/2.$$

Then:

$$M = (-B_R \cos t, 0, 0)$$
$$D = (0, A_R \sin t, -B_R \cos t)$$

and:

$$p_+ = M + D, \qquad p_- = M - D.$$

The nucleus is a derived midpoint coordinate, rather than a separately prescribed waveform.

BFS-20 [P] · Pole-to-nucleus noncontact

For either pole:

$$|p_\pm - M|^2 = A_R^2 \sin^2 t + B_R^2 \cos^2 t.$$

Both A_R and B_R are strictly positive for R>0.

Therefore:

$$|p_\pm - M| > 0$$

for every t.

Both poles remain separated from the relational midpoint. A common rigid frame transformation preserves this separation.

BFS-21 [P] · Midpoint covariance

For a shared affine frame transformation:

$$F(p) = o + Qp,$$

the two-point midpoint satisfies:

$$\frac{F(p_+) + F(p_-)}{2} = F(M)$$

Thus the midpoint construction survives translation and rotation of the entire pole pair.

BFS-21A [P] · Spatial pole decomposition under Shear-Fold

Let:

$$P_{\pm}(t) = o(t) + Q_S(t)p_{\pm}(t).$$

Their transformed nucleus is:

$$N(t) = o(t) + Q_S(t)M(t).$$

Then:

$$P_+(t) = N(t) + Q_S(t)D(t)$$
$$P_-(t) = N(t) - Q_S(t)D(t)$$

The two projected poles therefore remain symmetric about their transformed relational center.

This is an exact consequence of linearity, not a visual inference from the graphs.

D. Fold Progression and hierarchy

BFS-22 [D/P] · Fold Progression

Define linear anchor progression:

$$x_a(t) = \frac{2}{R}t.$$

For T=12π:

$$x_a(t+T) - x_a(t) = 24\pi/R.$$

Thus the internal fold can repeat relative to the advancing anchor while its absolute position continues translating.

BFS-23 [D] · Hierarchical frame definition

For each child:

$$P_i = P_{\text{parent}} + Q_{\text{parent}}w_i$$
$$Q_i = Q_{\text{parent}}Q_{\text{local},\,i}.$$

The transformations use the ordered multiplication convention of BFS-13.

BFS-23A [P] · Hierarchical matrix composition

For a hierarchy with local rotations Q_1, Q_2, …, Q_n:

$$H_n = Q_1Q_2\cdots Q_n$$

Every Q_i is orthogonal, so:

$$H_n^{\mathsf{T}}H_n = Q_n^{\mathsf{T}}\cdots Q_1^{\mathsf{T}}Q_1\cdots Q_n = I.$$

Thus inherited orientation remains orthogonal through any finite hierarchy.

This does not require the local rotations to commute.

BFS-24 [C] · Hierarchical closure theorem

Assume a finite frame hierarchy in which the root position, every local point coordinate, and every local frame rotation share period T.

For the root, recurrence holds by assumption.

If a parent position and orientation recur after T, and its child's local coordinates and rotation also recur, then the child's world-space position and orientation recur through the formulas of BFS-23.

By induction, recurrence extends to every descendant.

Therefore:

$$\text{Common component periodicity} \Rightarrow \text{hierarchical recurrence}$$

BFS-25 [C] · Reciprocal nested gearing

Interchanging the canonical angular rates 3/2 and 2/3 between nested levels preserves the 12π common phase recurrence.

Different ordered rotations can generate different trajectories, but the common recurrence remains when all components satisfy BFS-24.

BFS-26 [P] · N-object centroid covariance

For N ≥ 1 positions, write C_N for their centroid.

$$C_N = \frac{1}{N}\sum_{i=1}^{N} P_i.$$

Under a common affine frame F:

$$C_N\left(F(P_1), \ldots, F(P_N)\right) = F\left(C_N(P_1, \ldots, P_N)\right)$$

Writing each transformed point as $F(P_i) = o + QP_i$ gives the identity by distributing the sum. The centroid is a derived coordinate.

BFS-27 [C] · General winding commensurability

For reciprocal rates S/2 and 2/S, a finite common phase-return interval with a base rate of 1 exists when all three rates are rationally commensurate.

Every nonzero rational S satisfies this condition.

For arbitrary real S, common phase recurrence is not guaranteed.

E. Operator definitions and open correspondences

BFS-28 [D] · Fibron operator families

Wave-Fold: A local waveform carried by two or more rotation axes using a reciprocal pair of winding rates. In the two-axis base, the rates act on Y and Z. In the three-axis base, X shares the Z rate.

Shear-Fold: A three-axis reciprocal winding construction admitting paired opposing projections and their derived relational midpoint.

Fold Progression: Translation of a folded construction through an anchor coordinate.

Shear Division: A distinct amplitude-division and potentially multi-object centroid construction, reserved as a specialized extension.

Hierarchical Composition: Inheritance of orientation and location through nested frames.

BFS-29 [O] · Lens-to-winding assignment

The 120° reflected lens gives exact reciprocal phase-conversion factors 3/2 and 2/3.

Assigning those factors simultaneously to particular spatial rotation axes is a construction definition. The lens geometry alone does not establish a unique mandatory assignment.

BFS-30 [O] · Lens-to-waveform equivalence

The local trigonometric waveforms and the piecewise circular-lens boundary are not identical equations.

An explicit coordinate correspondence is required before declaring that a particular 3D projection equals the exact reflected lens.

BFS-31 [O] · Minimum spatial period and shape families

The 12π common phase recurrence does not establish that every projected trajectory has minimum period 12π or a specified number of independent geometric self-intersections.

Those are separate questions.

Mathematical scope

Banksian Fibron Space is a proposed geometric construction framework based on reciprocal winding, ordered rotational transformations, derived relational points, and hierarchical frame composition.

The register establishes the following results.

1. Exact circle and reflected-lens geometry.
2. Reciprocal geometric and winding identities.
3. Explicit two-axis and three-axis folding matrices.
4. Orthogonality, inverse transformations, and length preservation.
5. Exact common phase recurrence for the canonical winding configuration.
6. Exact axial pole, midpoint, and noncontact identities.
7. Recurrence through finite hierarchical frame composition.
8. The distinction between internal closure and translated world-space motion.

The framework's definitions do not themselves establish physical laws, unique natural necessity, or mathematical novelty over existing geometric formulations.